

Capstone Project Instructions- Step 6

Step 6: Building and Submitting Your Solution

Building and Testing:

Now that you have your preliminary Data Model, Control Flow, and API plan, start working on building your application.

Please follow the directions in the NextJS unit as a guidance for this step. NextJS framework will take care of building an always up-to-date barebones starter project.

<https://nextjs.org/docs/pages/api-reference/create-next-app>

Here's a list of steps that will help you plan this implementation.

- Configure project structure and folder organization using Next.js
- Implement CRUD operations for MongoDB database interactions.
- You are free to either develop a RESTful API to handle the data logic or use GraphQL or a combination of both if your project is “passing lots of data around”.
- Implement authentication and authorization mechanisms.
- Create UI components and layout using React.js.
- Integrate UI components with back-end APIs for data fetching and manipulation.
- Implement state management (Redux, Context API, or Equivalent).
- Style UI components using CSS..
- Conduct unit tests for individual components/modules and integration tests to validate end-to-end functionality.
- Debug and fix issues identified during testing.
- Deploy the application to Render or any other hosting platform.
- Create necessary documentation

Submission:

Before you submit make sure the submission includes Initial Project Ideas File, and Project Proposal Template.

You've done a lot of work so now it's time to show your mentor your progress! Create a README in markdown, make sure your GitHub is organized, and submit your finalized project that includes **all the steps of your Capstone project**.

Open a pull request from *dev* into *main* once you are ready to submit. Do **not** merge the pull request. Here is a [step-by-step guide](#) on how to create a pull request.

When you have completed this task, click Ready To Submit. <*READY TO SUBMIT button*>